

Programme Name/s	: Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Cloud Computing and Big Data/ Computer Technology/ Computer Engineering/ Computer Science & Engineering/ Data Sciences/ Computer Hardware & Maintenance/ Information Technology/ Computer Science & Information Technology/ Computer Science
Programme Code	: AI/ AN/ BD/ CM/ CO/ CW/ DS/ HA/ IF/ IH/ SE
Semester	: Fifth
Course Title	: OPERATING SYSTEM
Course Code	: 315319

I. RATIONALE

An Operating System is to manage a Computer Hardware and software resources efficiently and provide user friendly environment. An Operating System is a System Program that controls the execution of application program and acts as an interface between applications and the computer hardware. It also place a curtail role in maintaining system security, protecting data and ensuring that processes do not interfere with one another. This course enables to learn internal functioning of Operating System and will help in identifying appropriate Operating System for given Application/Task.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Interpret features of Operating System.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Explain the services and components of an Operating System.
- CO2 - Describe the different aspects of Process Management in an Operating System.
- CO3 - Implement various CPU Scheduling algorithms and evaluate their effectiveness.
- CO4 - Analyze the Memory Management techniques used by an Operating System.
- CO5 - Apply techniques for effective File Management in an Operating System.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme						Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH	Paper Duration		Theory				Based on LL & TL				Based on SL		Total Marks	
															Practical							
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA			
Max	Max	Max	Min	Max	Min	Max	Min	Max		Min	Max	Min										
315319	OPERATING SYSTEM	OSY	DSC	5	-	2	2	9	3	3	30	70	100	40	25	10	25@	10	25	10	175	

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 10 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe functions of an Operating System. TLO 1.2 Explain different services of Operating System. TLO 1.3 Explain use of system call of Operating System. TLO 1.4 Explain activities of Operating System in concern with their components.	Unit - I Operating System services and components 1.1 Operating System: concept, functions 1.2 Different types of Operating System: Batch Operating System, Multi-programmed, Time Shared Operating System, Multiprocessor System, Distributed System, Real Time System, Mobile OS (Android OS) 1.3 Command line based Operating System: DOS, UNIX GUI based Operating System: WINDOWS, LINUX, MaC OS 1.4 Different Services of Operating System, System Calls: Concept, types of system calls 1.5 Operating System Components: Process Management, Main Memory Management, File Management, IO Management, Secondary Storage Management	Presentations Lecture Using Chalk-Board
2	TLO 2.1 Explain the different states of a process. TLO 2.2 Describe the functions of different component of process stack in PCB (Process Control Block). TLO 2.3 Explain multiple processes access shared resources without interfering each other. TLO 2.4 Compare Multithreading models.	Unit - II Process Management 2.1 Processes: process state, process control block 2.2 Process Scheduling: scheduling queues, types of schedulers, context switch 2.3 Inter Process Communication: Shared memory system, Message passing system 2.4 Threads: Benefits, User and Kernel level threads, Multithreading Models: One to One, Many to One, Many to Many 2.5 Execute process commands like: top, ps, kill, wait, sleep, exit, nice	Lecture Using Chalk-Board Presentations

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Justify the need of given scheduling criteria with relevant example. TLO 3.2 Explain with example the procedure of allocating CPU to the given process. TLO 3.3 Calculate turnaround time and average waiting time of the given scheduling algorithm. TLO 3.4 Explain functioning of the given necessary conditions leading to Deadlock.	Unit - III CPU Scheduling 3.1 Scheduling: Basic concept, CPU and I/O burst cycle 3.2 Preemptive and Non-preemptive scheduling, scheduling criteria 3.3 Types of Scheduling algorithms: First Come First Serve(FCFS), Shortest Job First (SJF), Shortest Remaining Time Next (SRTN), Round Robin (RR), Priority Scheduling, Multilevel Queue Scheduling 3.4 Deadlock: System Models, Necessary conditions Leading to Deadlock, Deadlock Handling: Deadlock prevention, Deadlock avoidance- Banker's Algorithm	Presentations Lecture Using Chalk-Board
4	TLO 4.1 Compare fixed and variable memory partitioning. TLO 4.2 Differentiate between Bit map and Linked list technique. TLO 4.3 Explain working of various partitioning algorithm. TLO 4.4 Calculate page fault for given page reference string.	Unit - IV Memory Management 4.1 Basic Memory Management: Partitioning - Fixed and Variable, Free Space Management Techniques: Bit map, Linked List 4.2 Swapping, Compaction, Fragmentation, Partitioning Algorithms: First fit, Best fit, Worst fit 4.3 Non-contiguous Memory Management Techniques: Paging, Segmentation 4.4 Virtual Memory: Basics, Demand paging, Page Fault 4.5 Page Replacement Algorithm: First In First Out (FIFO), Least Recently Used (LRU), Optimal	Lecture Using Chalk-Board Presentations Video Demonstrations
5	TLO 5.1 Explain structure of the given file system with example. TLO 5.2 Describe mechanism of file access method. TLO 5.3 Explain procedure to create access directories and assign the given file access permissions.	Unit - V File Management 5.1 File Concepts: Attributes, Operations, File types and File system structure 5.2 Accessing Methods: Sequential, Direct 5.3 File Allocation Methods: Contiguous allocation, Linked allocation, Indexed allocation 5.4 Directory Structure: Single level, Two level, Tree structured Directory	Presentations Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Execute the system call commands.	1	* System call commands in Linux such as fork(), exec(), getpid, pipe, exit, open, close, stat, uname.	2	CO1
LLO 2.1 Execute process related commands.	2	* Process related commands in Linux - top, ps, kill, wait, sleep, nice, renice, bg, fg.	2	CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Execute message passing and shared memory commands.	3	* a. Commands for Sending Messages to Logged-in Users -who, cat, wall, write, mesg. * b. List Processes Attached to a Shared Memory Segment: ipcs.	2	CO2
LLO 4.1 Implement First Come First Serve (FCFS) Scheduling algorithm.	4	* Write a C/Python program to calculate average waiting time and Turnaround Time of n processes with First Come First Serve (FCFS) CPU scheduling algorithm.	2	CO3
LLO 5.1 Implement Shortest Job First (SJF) Scheduling algorithm.	5	Write a C/Python program to calculate average waiting time and Turnaround Time of n processes with Shortest Job First (SJF) CPU scheduling algorithm.	2	CO3
LLO 6.1 Implement Priority Scheduling algorithm.	6	Write a C/Python program to calculate average waiting time and Turnaround Time of n processes with Priority CPU scheduling algorithm.	2	CO3
LLO 7.1 Implement Round Robin (RR) Scheduling algorithm.	7	Write a C/Python program to calculate average waiting time and Turnaround Time of n processes with Round Robin (RR) CPU scheduling algorithm.	2	CO3
LLO 8.1 Implement Banker's algorithm for deadlock avoidance.	8	Write a C/Python program to implement Banker's Algorithm.	2	CO3
LLO 9.1 Execute memory management commands.	9	Basic memory management commands - df, free, vmstat, /proc/meminfo, htop.	2	CO4
LLO 10.1 Implement First In First Out (FIFO) Page Replacement algorithm .	10	* Write a C/Python program on First In First Out (FIFO) Page Replacement algorithm.	2	CO4
LLO 11.1 Implement Least Recently Used (LRU) Page Replacement algorithm.	11	Write a C/Python program on Least Recently Used (LRU) Page Replacement algorithm.	2	CO4
LLO 12.1 Implement sequential file allocation method.	12	* Write a C/Python program on sequential file allocation method.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)**Assignment**

- Find out the total number of page faults using – i) First In First Out ii) Least recently used page replacement ii) Optimal page replacement Page replacement algorithms of memory management, if the page are coming in the order 7

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0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

- Compare between CLI based Operating System and GUI based Operating System.
- Differentiate between process and thread (any two points). Also discuss the benefits of multithreaded programming.
- Enlist different file allocation methods? Explain contiguous and indexed allocation method in detail.

Micro project

- Create a report depicting features of different types of operating systems- Batch operating system, Multi programmed, Time shared, Multiprocessor systems, Real time systems, Mobile OS with examples.
- Implement and Compare Memory Allocation Strategies - First Fit, Best Fit, Worst Fit
- Create a report on different operating system tools used to perform various functions.

Self learning

- Complete any one course related to the operating system on MOOCS such as NPTEL, Coursera, Infosys Springboard etc.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Computer system with basic configuration. Linux or alike operating system such as Ubuntu, CentOS or any other.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Operating System services and components	CO1	10	2	8	4	14
2	II	Process Management	CO2	10	4	4	6	14
3	III	CPU Scheduling	CO3	10	2	6	8	16
4	IV	Memory Management	CO4	12	2	6	8	16
5	V	File Management	CO5	8	2	4	4	10
Grand Total				50	12	28	30	70

X. ASSESSMENT METHODOLOGIES/TOOLS

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- Continuous assessment based on process and product related performance indicators. Each practical will be assessed considering 1) 60% weightage is to process 2) 40% weightage to product

Summative Assessment (Assessment of Learning)

- End Semester Examination, Lab Performance, Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	-	-	2	-	-	1			
CO2	1	-	-	2	1	-	-			
CO3	1	1	1	2	1	-	-			
CO4	2	2	2	2	1	-	2			
CO5	2	2	2	2	1	-	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
 *PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Dhananjay M. Dhamdhare	Operating System: A Concept-Based Approach	McGraw Hill Education 3rd edition, ISBN: 978-1259005589
2	William Stallings	Operating Systems : Internals and Design Principles	Pearson Education 9th Edition, ISBN: 978-9352866717
3	Richard Petersen	Linux The Complete Reference	McGraw Hill, 6th edition, ISBN: 978-0071492478
4	Richard Blum	Linux command line and shell scripting	Wiley India, ISBN: 978-1118983843
5	Abraham Silberschatz and James Peterson	Operating System Concepts	Wiley India, ISBN: 9781119454083

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://archive.nptel.ac.in/courses/106/105/106105214/	Introduction to Operating System
2	https://www.geeksforgeeks.org/processes-in-linuxunix/	Process Related commands
3	https://ubuntu.com/download/desktop	Installation of Ubuntu
4	https://developers.redhat.com/products/rhel/download	RedHat Linux download

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Sr.No	Link / Portal	Description
5	https://www.digitalocean.com/community/tutorials/linux-commands	Basic Linux commands
6	https://www.geeksforgeeks.org/what-is-an-operating-system/	Operating System
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 24/02/2025**Semester - 5, K Scheme**